Research, Innovation, Incubation & Commercialization towards Building Developed Bangladesh (Sonar-Bangla)

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AIMS Lab
Advanced Intelligent Multidisciplinary Systems Lab

CMED
Connecting People, Saving Lives

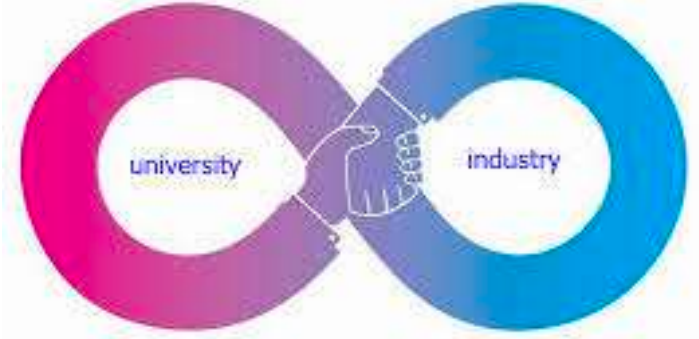
Where We Live: Bangladesh



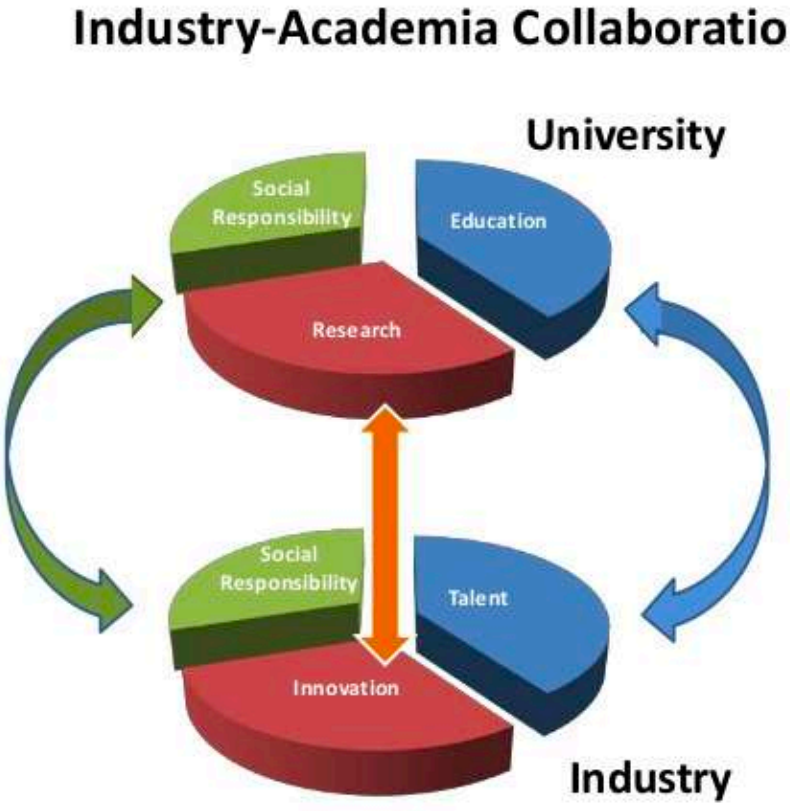


Infrastructure of Digital Bangladesh

63.5% internet Growth



Bangladesh : What We Need & Why



Faculty Development Program & Commitment to Society



Infrastructure & facilities for research

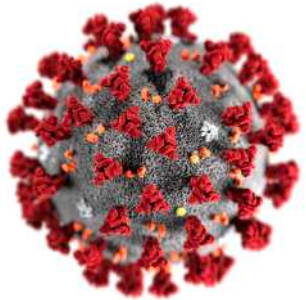
Capacity to solve social problems





Quickly adapting the Situations

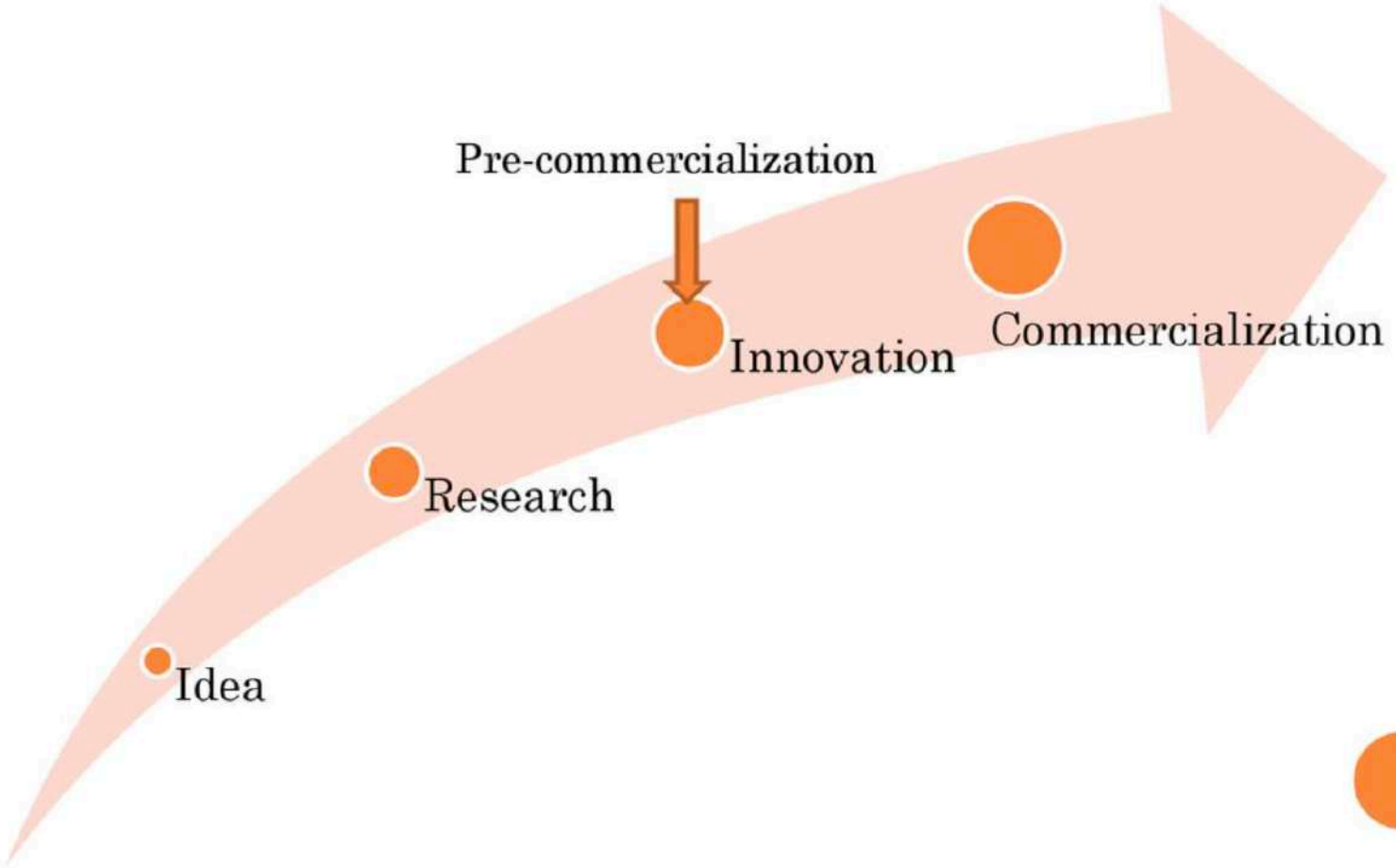
COVID-19 is one of the example





Research, Innovation, Incubation, Commercialization

RESEARCH-INNOVATION-COMMERCIALIZATION – GENERAL MODEL

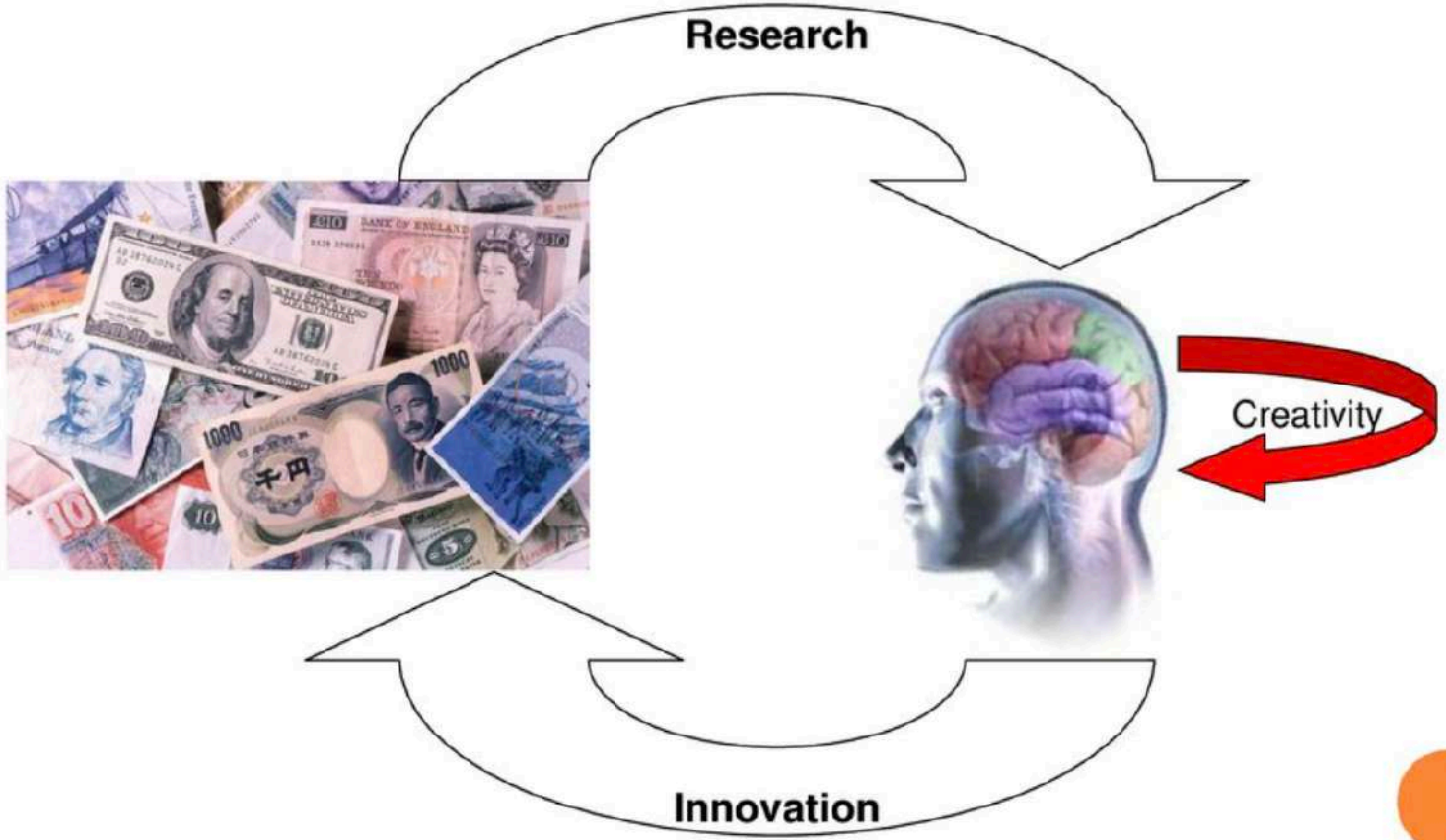



idea

WHAT IS RESEARCH?

- **Research** can be defined as the search for knowledge or as any systematic investigation, with an open mind, to establish novel facts, solve new or existing problems, prove new ideas, or develop new products/services/process

Research vs Innovation – What is the relationship?



Definition of Innovation

Innovation = Invention*Commercialization

The Differences

Innovation

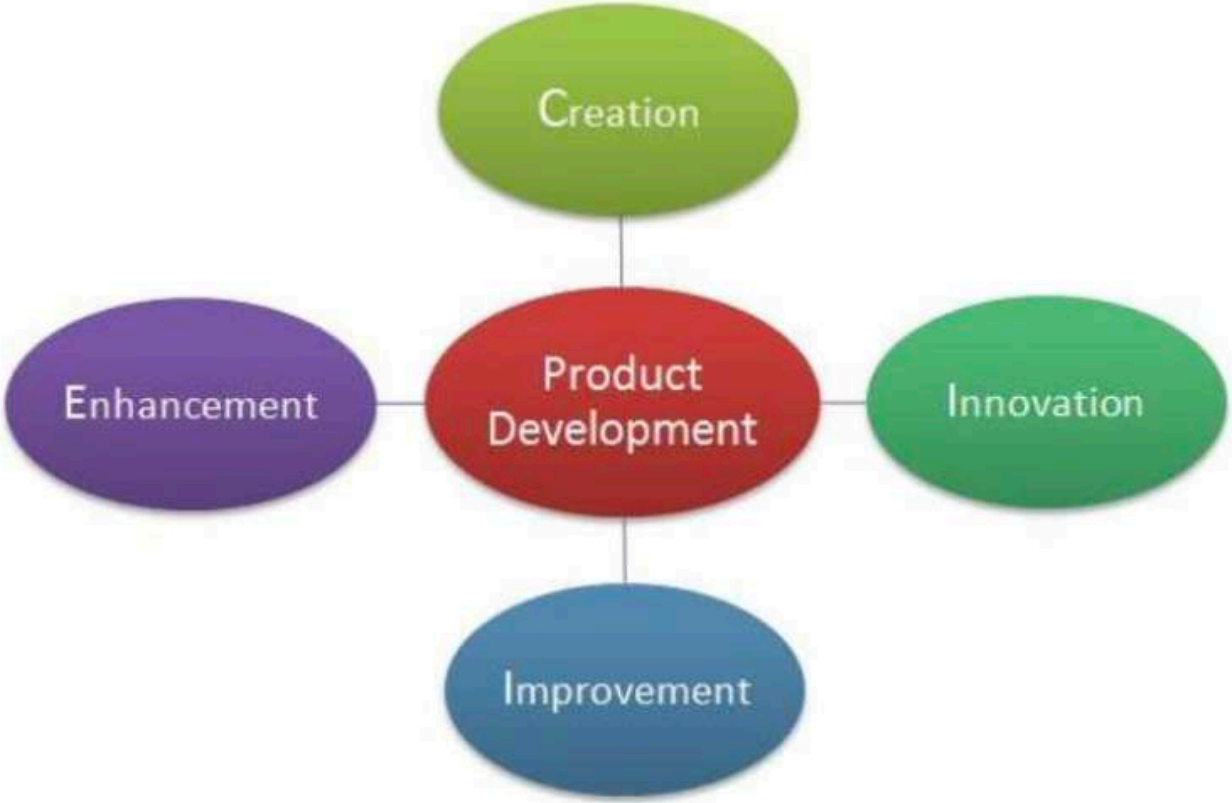
- Innovation happens when someone "improves on or makes a significant contribution" to something that has already been invented. Steve Jobs was an innovator

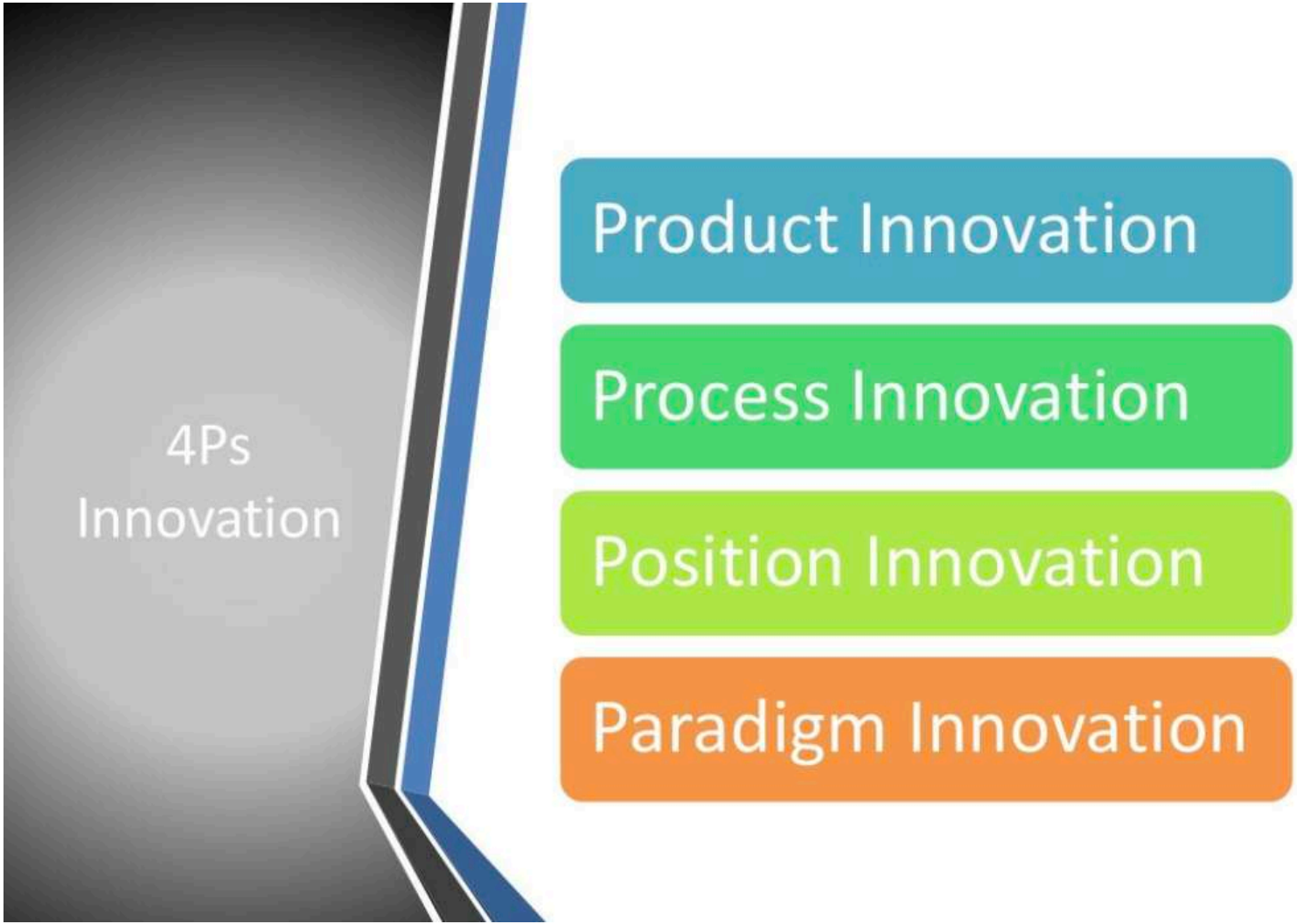
Invention

Invention is the "creation of a product or introduction of a process for the first time." Thomas Edison was an inventor.

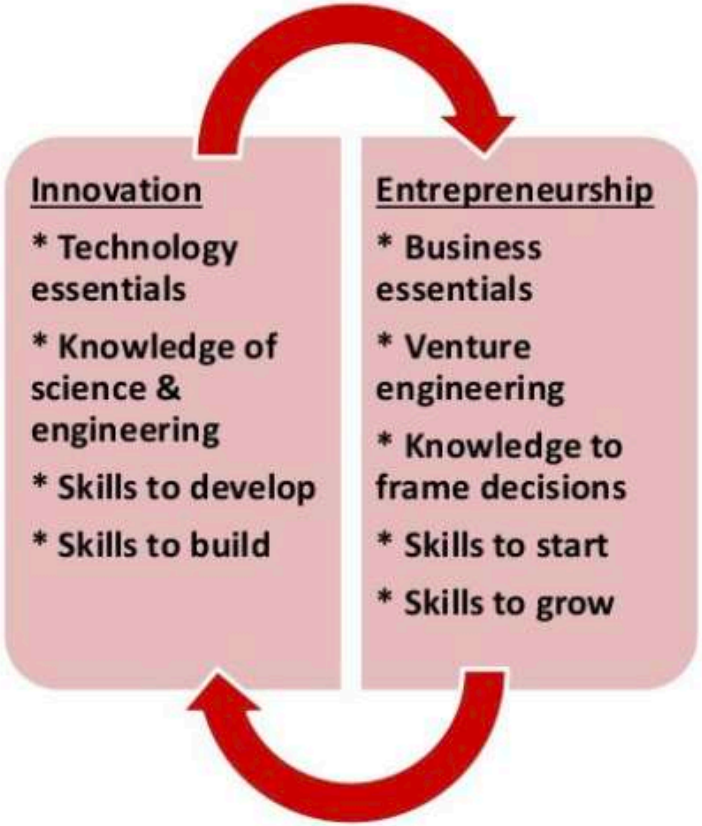
(Bhasin, K., 2012)

Innovation


Creation
Enhancement
Product Development
Innovation
Improvement


4Ps
Innovation
Product Innovation
Process Innovation
Position Innovation
Paradigm Innovation

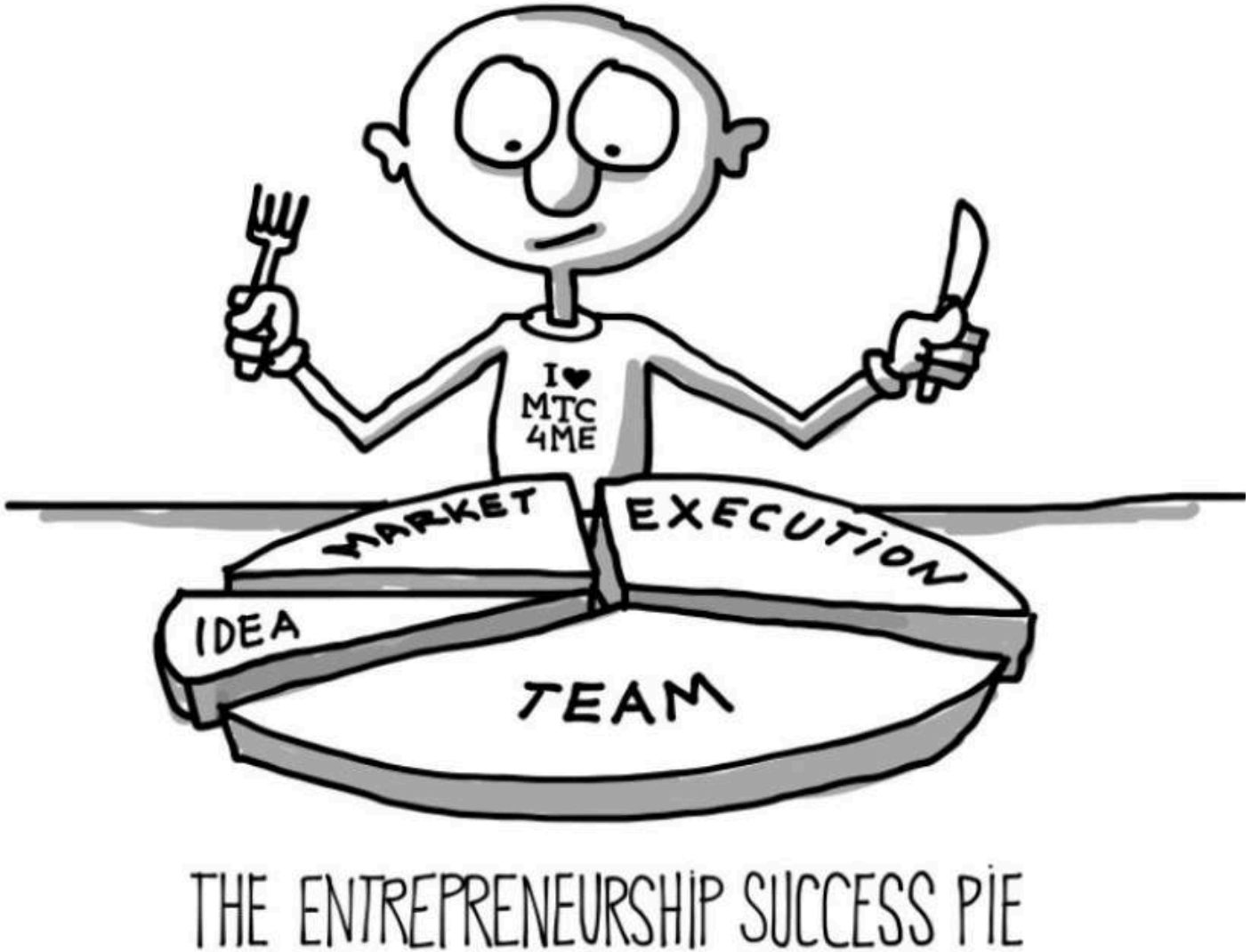
What Is Entrepreneurship?



Successful Entrepreneurship



Spirit + Skills



THE ENTREPRENEURSHIP SUCCESS PIE

New Product Development

- What is the gap in the market?
- What is the trend?
- What is the new product about?
- Who needs the new product?
- How much does it cost?
- Is there enough market to recover the cost?
- Is there any intelligent report as evidence?
- Who are the competitors?



By: Oguchi Martins Egbujor

Intellectual Property Tools

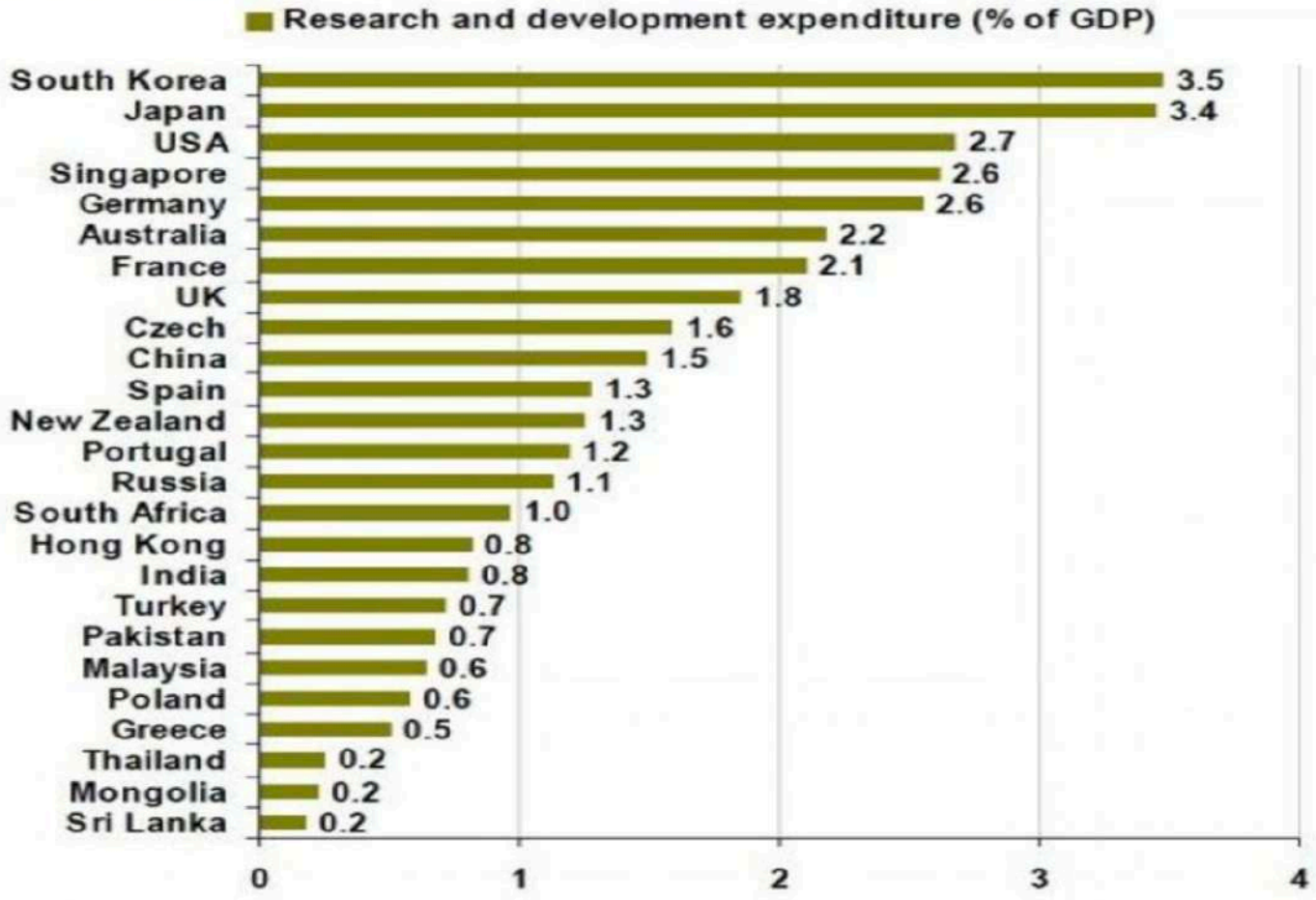
- Copyrights
- Patent
- Trade Mark
- A design

Ownership of Intellectual Property

You own intellectual property if you:

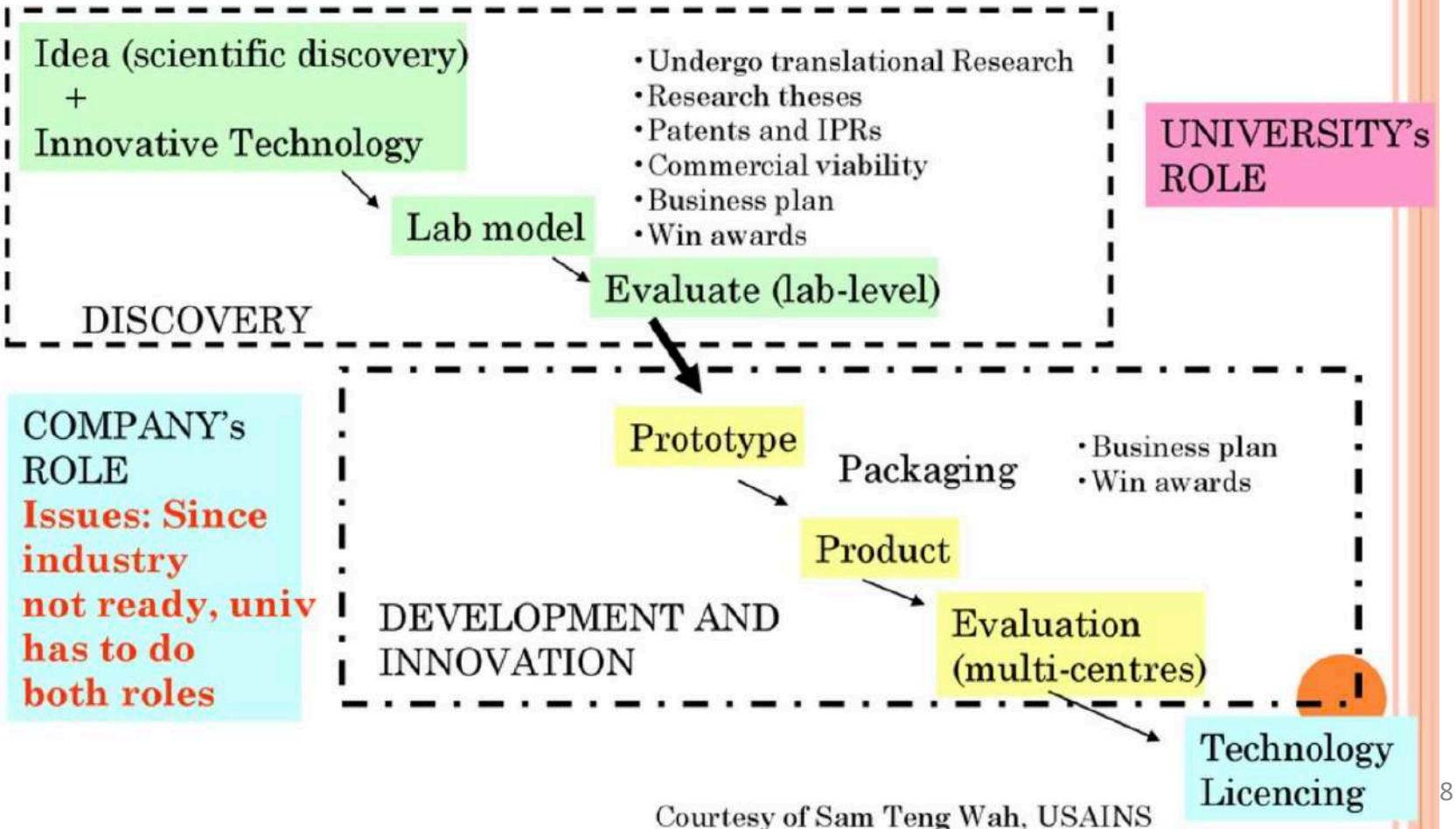
- Created it (and it meets the requirements for copyright, a patent or a design)
- Bought intellectual property rights from the creator or a previous owner
- Have a brand that could be a trade mark, eg a well-known product name
- Intellectual property can:
- have more than one owner
- belong to people or businesses
- be sold or transferred

Chart 5: R&D expenditure as % of GDP (%)



Source: World Bank. Expenditures for research and development are current and capital expenditures (both public and private) on creative work undertaken systematically to increase knowledge, including knowledge of humanity, culture, and society, and the use of knowledge for new applications. R&D covers basic research, applied research, and experimental development.

Commercialization of Knowledge



Courtesy of Sam Teng Wah, USAINS

A startup timeline



Ideation

Potential scalable product/service idea for big enough target market. Some initial revenue models for how it would make money. One person OR only vague team; no confirmed commitment and/or no right skills balance in the team structure yet

Concepting

Having clear and meaningful target with clear direction for min. 3 years with milestones to get there, -> 3, 6, 12, 24, 36 months... Having team of two or three core founding people with balanced ownership. Can also already have some extended team with lighter commitment (stock options and/or cash compensation)

Commitment

Committed & skills balanced founding team. Able to develop the product/service (Minimum Viable Product) without dependency of uncommitted external resources OR already have initial product/ service developed. Have signed shareholder agreement between founders, with milestones, committed time and money usage, for min. 2+ years with vesting etc.

Validation

Can already show some user growth and/or revenue (initial traction). AND/OR continue to attract additional resources (money or sweat equity) for equity or future revenues. Looking for clear market validation (Product Market Fit), to be able to move into scaling.

Scaling

Showing clear, growing and measurable user/market traction in big or rapidly growing target market. Can and want to scale fast. AND/OR is able to attract significant funding.

Establishing

Achieved great growth, that can expected to continue strong. No longer need to "try" get resources and can get those easily. Continue to grow and often wants to culturally continue behaving like a "startup" for as long as possible. Founders make exit or continue biz as usual.

Startup incubators

- **incubator**: Program designed to support successful development of startups by providing business resources.
  - business, marketing, and networking advice
  - computing resources (computers, net access, servers)
  - financial advisors
  - management teams
  - access to Investment, loans and banks
  - access to angel investors and/or venture capitalists
  - legal advisors (such as for intellectual property rights)
  - Examples: GP Accelerator, Ycombinator, Techstars

GP ACCELERATOR

Y Combinator

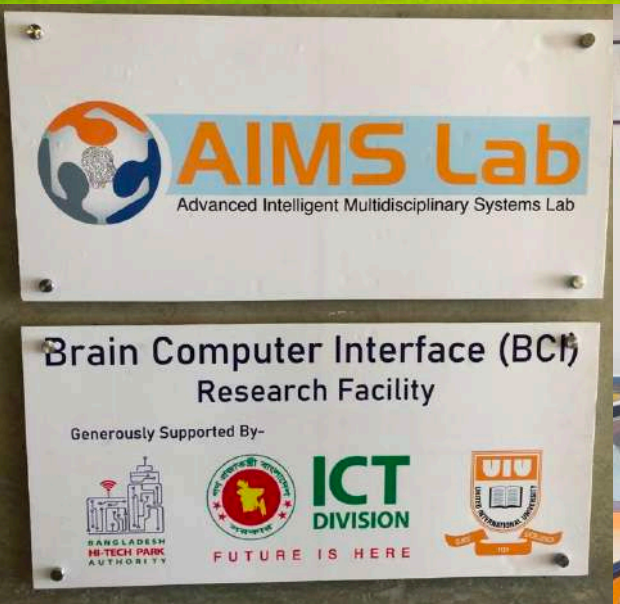
techstars



https://www.ted.com/talks/bill_gross_the_single_biggest_reason_why_start_ups_succeed?language=en#t-12216

AIMS Lab
Advanced Intelligent Multidisciplinary Systems Lab







CMED: Providing villagers with affordable healthcare at home through ‘health accounts’

This digital health tech startup is using AI-driven health kits and trained paramedics to serve the rural and semi-urban population

Regular User : 1.3M
COVID 19 User: 2M
Current Valuation: $ 5M



CMED’s trained paramedics go door to door documenting health conditions like blood pressure, diabetes, BMI etc. Photo: Courtesy

A phone call from Bangladesh informed Professor Khondoker Abdullah Al Mamun of his mother's fourth stage breast cancer diagnosis. At the time, he was doing his PhD at the University of Southampton, United Kingdom.

AIMS LAB, UNITED INTERNATIONAL UNIVERSITY ACHIEVED 4 AWARDS OUT OF TOP 10 IN MUJIB100 IDEA CONTEST 2021



AI Customer Service: The Future Smart Customer Support for Better Customer Satisfaction



PVDoctor: Smartphone-based Virtual tool for Parkinson's Disease Identification and Monitoring



DEEP-DEPRESSION: Smart Diagnostic Tool for Early Identification of Depression Biomarkers



iPower: An Intelligent Electricity Demand Forecast and Optimization Application

The projects were selected out of a total 1100+ ideas.Award amount is BDT 40,00,000.00

AIMS Lab
Advanced Intelligent Multidisciplinary Systems Lab

MUJIB 100
IDEA CONTEST 2021

UNITED INTERNATIONAL UNIVERSITY



ব্রেইন কম্পিউটার ইন্টারফেস (বিসিআই) গবেষণাগার

Brain Computer Interface (BCI) Research Lab

United International University

শুভ উদ্বোধন

জনাব জুনাইদ আহ্মেদ পলক এমপি

মাননীয় প্রতিমন্ত্রী, তথ্য ও যোগাযোগ প্রযুক্তি বিভাগ

১২ ডিসেম্বর ২০২১, ২৫ অগ্রহায়ন ১৪২৮

বাস্তবায়নেঃ কালিয়াকৈর হাই-টেক পার্ক (এবং অন্যান্য হাই-টেক পার্ক)-এর উন্নয়ন প্রকল্প,
বাংলাদেশ হাই-টেক পার্ক কর্তৃপক্ষ, তথ্য ও যোগাযোগ প্রযুক্তি বিভাগ

DIGITAL BANGLADESH
Skilled. Equipped. DigitalReady.

ICT DIVISION
FUTURE IS HERE

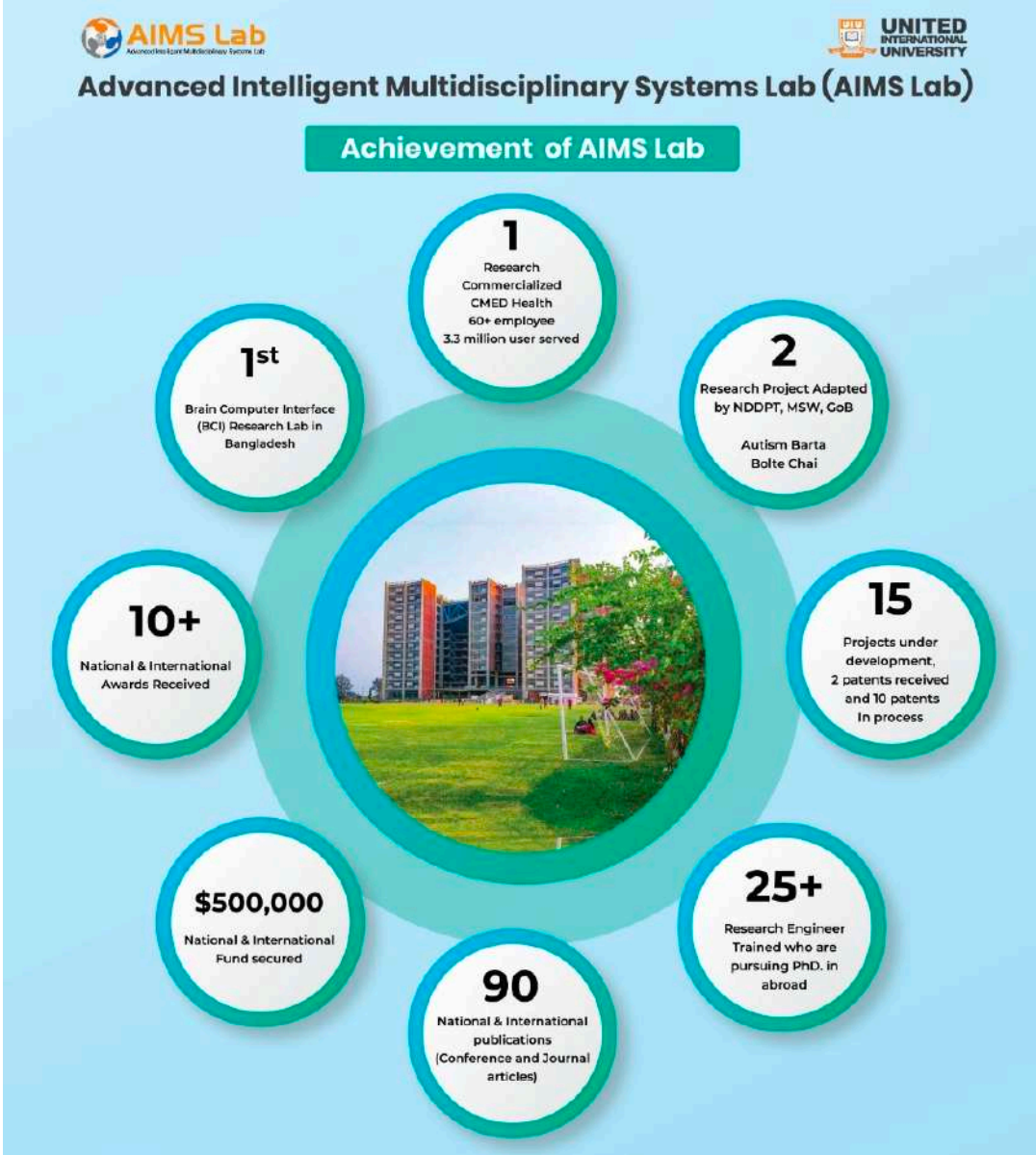
BANGLADESH HI-TECH PARK AUTHORITY

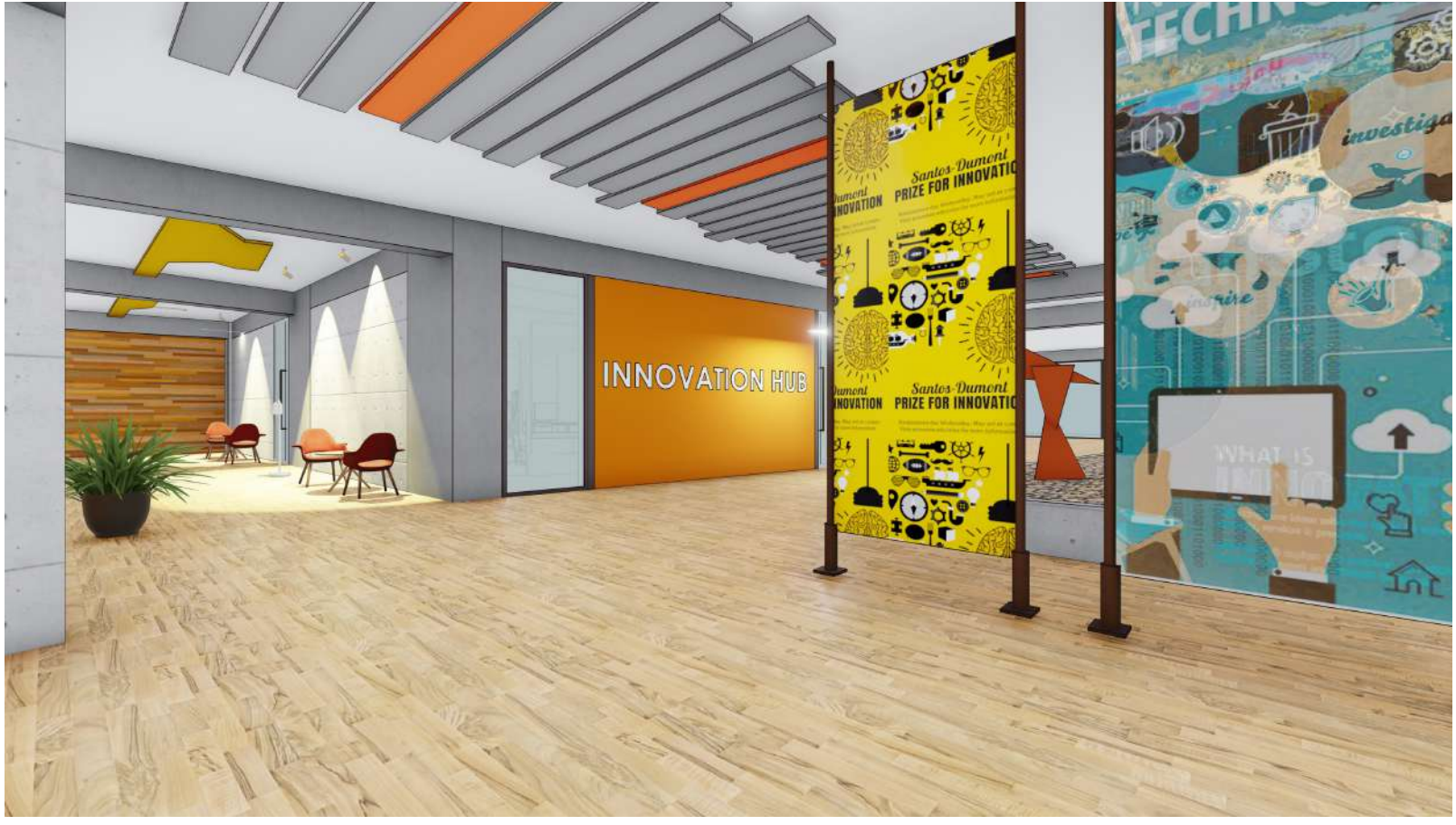
UNITED INTERNATIONAL UNIVERSITY

Accelerating Research Innovation Incubation & Commercialization Ecosystem in Bangladesh

AIMS Lab
Advanced Intelligent Multidisciplinary Systems Lab

UNITED INTERNATIONAL UNIVERSITY
QUEST FOR EXCELLENCE





UIU University Innovation Hub

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UIU University Innovation Hub

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